

NewsMatrix Detailed Features

This document provides a comprehensive breakdown of all features in **NewsMatrix**, including access control, public features, journalist workflows, admin portals, and authentication systems.

1. Access Control (Role-Based Access Control - RBAC)

The security model is implemented consistently across the stack:

- **Frontend (Vue Router Guards):** Prevents unauthorized route navigation based on roles stored in the active session.
- **Backend (FastAPI Dependencies):** Inspects JWT claims in the incoming `Authorization` header. If authorization fails, it returns a `403 Forbidden` error.

Roles:

- **Admin (Administrator):** Manages user accounts, configures news organizations, structures article categories, and uploads documents for RAG processing.
 - **Journalist:** Manages articles belonging to their specific organization. Permissions include drafting, publishing, editing, and deleting articles.
 - **User (Regular User):** Reads public news, likes articles, comments on published pieces, follows news agencies, and sends messages via Inbox.
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2. Public Client Features

2.1. Homepage

- Styled as a premium Bento Box layout, linking to the News Feed, Organizations directory, and authentication portals.
- Integrates an interactive photo carousel with responsive preview effects.

2.2. News Feed

- **Published News Feed:** Restricts listing to `Published` articles, hiding drafts. Cards display titles, summaries, lists of authors, categories, and interaction counters (likes count, comments count, followers count).
- **Search & Filters:**
 - Full-text search across titles, contents, and categories.
 - Quick-filters by category selection.
 - Date range filters to retrieve articles published on specific days.
- **Multi-source Toggle:** Users can switch the data source dynamically:
 - *System data:* Real production database data served via Backend APIs.

- *Mock data*: Static mock data read from local JSON files (`news.json`).
- *External data*: Calls GNews API. Implements a 60-second caching mechanism and customizable API URLs.
- **Pagination**: User-controlled page sizes (5 or 10 articles) with easy page-flipping controls.
- **Feed Interactions**: Logged-in users can like/unlike and follow/unfollow organizations directly from the feed card.

2.3. News Detail Page

- Displays full content alongside author details, publication date, and category pills.
- **Asynchronous Comments**:
 - Chronological list of user comments with sender emails and timestamps.
 - Post comments asynchronously using RabbitMQ queues to guarantee high throughput and minimize blocking.
- **Likes & Follows**: Real-time updates for like counters and followed news agencies.

2.4. About Page

- Details core mission goals and development roadmap milestones.
 - Toggles an interactive Vue sandbox demonstrating Two-Way Data Binding using user inputs.
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3. Journalist Workspace

Journalists are greeted with a dedicated workspace dashboard upon signing in:

- **Organization Catalog**: View all articles created by the journalist's organization (both Draft and Published).
 - **Create Article**: Form inputs for title, content, status selection, and category tagging.
 - **Edit Article**: Update existing articles. The system validates the organization's edit quota (`current_edit_limit`) to restrict excessive edits.
 - **Delete Article**: Permanently delete articles after confirming with a modal check.
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4. Admin Portal

4.1. Organization Management

- **Overview Dashboard**: List news organizations, showing daily limits (`daily_post_limit`), editing limits (`current_edit_limit`), and total followers.
- **Create/Update Organizations**: Initialize or configure news organizations.

- **Staff Management (Organization Detail Page):**
 - List journalists working under the organization (paginated at 5 per page).
 - Add free-agent journalists to the organization.
 - Remove (dismiss) journalists from the organization.

4.2. Category Management

- List all categories alphabetically.
- Create new categories (guarantees unique, non-empty names).
- Update category names and delete unused categories.

4.3. User Governance

- Bento Box visual metrics tracking user counts and active roles.
- Detailed user list showing emails, assigned roles (colored badges), and updates.
- Register new accounts, update roles, reset passwords, or delete accounts.

4.4. Document Ingestion

- **PDF Upload:** Admin uploads knowledge base PDF documents.
 - **Auto-Pipeline:** Triggers Supabase Storage transfers, chunking, and database indexing.
 - **Ingestion Log:** Monitors ingested PDFs, showing page counts, chunk counts, files formats, paths, and timestamps.
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5. User Authentication & Security

- **User Sign Up:** Account creation with email and password, default role set to **User**.
- **Standard Sign In:** Standard credential check, returning a JWT token for authorization headers.
- **Social Sign In (OAuth2):** Login via Google or GitHub. If the account email is not in the system, it automatically registers a new account under the **User** role.
- **Logout:** Clears token storage and redirects to the login screen.